

5-5-1: Examining Load Balancers and Proxies

At the end of this episode, I will be able to:

1. Identify and explain the fundamental role and functionality of load balancers and proxies, given a scenario.

Learner Objective: *Identify and explain the fundamental role and functionality of load balancers and proxies, given a scenario*

Description: In this episode, the learner will examine load balancers and proxies. We will explore various types, services, and functionalities provided by these network components.

- What are load balancers used for on networks?
 - o Distribute network or application traffic across multiple servers
- What is the purpose of traffic distribution?
 - o Improves responsiveness
 - o Increases availability of applications, websites, or databases.
- What are some of the different types of load balancers?
 - o Round Robin - distributes requests sequentially across a group of servers in a repeating pattern.
 - o Least connection - routes traffic to servers with the fewest active connections, useful in situations where server load is not evenly distributed.
 - o Weighted connection - a weight is assigned to each server, higher weight values indicating a greater request capacity
 - o IP hash - This method uses the client's IP address or an HTTP header value to determine which server receives the request, ensuring that users consistently connect to the same server.
 - o Application-aware - makes routing decisions based on application-specific data such as HTTP headers, cookies, or data within the message content.
- What are proxies?
 - o An intermediary for incoming clients request for resources from other servers
- What is the purpose of proxies?
 - o Provides caching services
 - ♣ Stores copies of frequently requested web resources.
 - ♣ Delivers cached resources directly, with no need for external data fetching.
 - ♣ Reduces latency and bandwidth consumption
 - ♣ Implements cache control policies for storage management.
 - o Provides security
 - ♣ Acts as an intermediary, intercepting requests to external servers.
 - ♣ Mitigates direct external attacks, filtering malicious traffic.
 - ♣ Enforces security policies like access control and content filtering.
 - o Anonymization services
 - ♣ Makes requests on behalf of the client, exposing only the proxy's IP address.
 - ♣ Useful in scenarios for bypassing geo-restrictions and maintaining privacy online.
- What are different proxies types?
 - o Forward proxy - sits between the client and the Internet to control and filter client requests, enhance security, or bypass restrictions.
 - o Reverse proxy - sits in front of web servers, forwarding client requests to those web servers, used for load balancing or caching.
 - o Transparent proxy - intercepts communication at the network layer, not requiring special client configuration.
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- Web application proxies - sit in front of web servers, forwarding client requests to internal applications for external use (on-premise to mobile, cloud, hybrid...etc.), which is another purpose of proxies; also providing security services.

- Additional Resources:
 - o If applicable